

Why do protists make poor mathematicians? Each question is worth 2 points except where noted. Good luck!

Match the *most appropriate* term to each concept. Not all terms are used. No term is used twice.

Active Diffusion	Gametophyte	Pseudopodia
Amplification	Gastrovascular Cavity	Reception
Binary Fission	Haplogamy	Response
Chemotaxis	Karyogamy	Run and Tumble
Cytokinesis	Magnification	Second Messenger
Cytoplasmic Streaming	Mechanotaxis	Signalation
Cytotaxis	Meiosis	Sporophyte
Diplogamy	Mitosis	Syngamy
Dive and Bomb	Netflix and Chill	Thigmotaxis
Drive and Orient	Passive Diffusion	Touchyfeelytaxis
Galvanotaxis	Plasmosyngamy	Transduction

- 1. Gametophyte** Mosses spend most of their life in this life cycle stage.
- 2. Amplification** Increases the strength of a weak signal.
- 3. Plasmosyngamy** Life cycle stage where fungal cells have fused but nuclei have not.
- 4. Transduction** A signal is transmitted to a target area inside a cell.
- 5. Cytokinesis** Division of the cell after mitosis.
- 6. Run and Tumble** Allows bacteria to move towards a chemical cue.
- 7. Second Messenger** A receptor activates these ions inside a cell.
- 8. Cytoplasmic Streaming** Transports nutrients throughout a cell.
- 9. Thigmotaxis** Locomotion in response to touch.
- 10. Gastrovascular Cavity** Responsible for digestion and nutrient distribution in flatworms.
- 11. ANY ANSWER** Name the band that you think is Taylor's favorite (not in list above).

Multiple choice. Choose the *best* answer. Clearly circle the letter for your answer. Vague or unclear marks will be counted as incorrect.

12. The protein that is activated in response to auxin that allows phototropism to occur is
- A. proton channel.
 - B. basal apparatus.
 - C. abscisic acid.
 - D. transmembrane.
 - E. expansin.**
 - F. wallulase.
13. Which of the following about phloem is false?
- A. The cell has cytoplasm.
 - B. Sugars move by passive diffusion.
 - C. Sieve-tube elements do not have a nucleus.
 - D. The plasma membrane is modified to form a sieve plate.**
 - E. Companion cells provide proteins to sieve-tube elements.
14. During protist conjugation, which of the following does *not* occur during some part of the process?
- A. Cells exchange micronuclei.
 - B. The macronucleus fuses with the micronucleus.**
 - C. The micronucleus undergoes meiosis.
 - D. The micronucleus undergoes mitosis.
 - E. Some micronuclei distintegrate.
 - F. The macronucleus distintegrates.
15. When ATP attaches to the myosin head
- A. the myosin head attaches to the actin filament.
 - B. the actin head attaches to the myosin filament.
 - C. the myosin head releases from the actin filament.**
 - D. the actin head attaches to the myosin filament.
 - E. the myosin head initiates the power stroke.
16. Which of the following are primarily responsible for movement in the eukaryotic flagellum?
- A. Hydrogen ions and basal apparatus.
 - B. Actin and myosin filaments.
 - C. Basal apparatus and hook.
 - D. Dynein and microtubules.**
 - E. Sensory and motor neurons.
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17. The fluid that moves around in the circulatory system of a typical arthropod is the
- A. cytoplasm.
 - B. blood plasma.
 - C. hemolymph.**
 - D. gastrovascular fluid.
 - E. intracellular fluid.
18. Ignoring all other factors, what kind of day would result in the fastest delivery of water and minerals to the leaves of a tree?
- A. A cool, dry day.
 - B. A cool, humid day.
 - C. A very hot, dry, windy day.
 - D. A warm, dry day.**
 - E. A warm, humid day.
19. The main stimulus used by birds to find their way north for the summer time is
- A. thermal.
 - B. mechanical.
 - C. electromagnetic.**
 - D. thigmic.
 - E. chemical.
20. Most of the water taken up by a plant
- A. transports sugars in the phloem.
 - B. is lost during transpiration.**
 - C. is converted to sugars during photosynthesis.
 - D. enters the central vacuoles of cells.
 - E. keep guard cells turgid.
21. The highest quality form of energy (best able to do work) for organisms is
- A. kinetic.
 - B. thermal.
 - C. radiant.**
 - D. chemical.
 - E. potential.
22. Transpiration in plants requires all of the following *except*
- A. adhesion of water molecules to cellulose.
 - B. evaporation of water molecules.
 - C. cohesion between water molecules.
 - D. active transport through xylem cells.**
 - E. water vapor on the spongy mesophyll.
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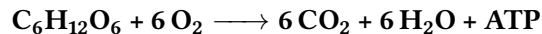
23. Organisms with a circulating fluid that is separate from the fluid that directly surrounding the body's cells are likely to have
- A. a gastrovascular cavity.
 - B. an open circulatory system.
 - C. a hemolymph system.
 - D. a closed circulatory system.**
 - E. a vascular system.
24. Differences in energy quality and availability affects the amount of surface area and volume in organisms. Compared to animals, plants have
- A. high surface area and low volumes.**
 - B. low surface area and high volumes.
 - C. low surface areas and low volumes.
 - D. high surface area and high volumes.
25. What drives the flow of water through the xylem?
- A. The evaporation of water from the leaves.**
 - B. Absciscic acid opening guard cells around the stomata.
 - C. The number of companion cells in the phloem.
 - D. Transpirational pull through sieve-tube elements.
 - E. A higher concentration of sugars in the leaves than in the roots.
26. The type of nerve cell responsible for integrating different stimuli is the
- A. interneuron**
 - B. motor neuron
 - C. dendrite
 - D. sensory neuron
 - E. axon
27. In terms of structure, cilia are most like
- A. microtubules.
 - B. bacterial flagella.
 - C. myosin filaments.
 - D. actin filaments.
 - E. eukaryotic flagella.**
28. The ion used most often as a second messenger is
- A. K^+
 - B. H^+
 - C. Ca^{+2}**
 - D. Cl^-
 - E. Na^+
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29. From an energy perspective, animals are mobile because they consume
- A. low quality energy and carbon intake is constant.
 - B. low quality energy and low energy efficiency.
 - C. high quality energy and carbon intake is irregular.
 - D. high quality energy and carbon intake is constant.
 - E. low quality energy and carbon intake is irregular.**
 - F. high quality energy and high energy efficiency.
30. The function of valves in veins in a closed circulatory system is to
- A. provide support for the extra muscles around the vein.
 - B. enhance diffusion of nutrients to the tissues.
 - C. decrease the amount of pressure in the veins.
 - D. enable the vein to withstand high pressure.
 - E. ensure one way flow of circulatory fluids.**
31. The vascular tissue system for delivery of water, minerals, and nutrients throughout the plant includes all of the following except for
- A. xylem.
 - B. phloem.
 - C. sieve plates.
 - D. cohesion.**
 - E. companion cells.
32. Hydrostatic skeletons are
- A. skeletons made of microtubules in the cytoplasm of protists.
 - B. skeletons found in deep sea organisms that depend on high pressure of the surrounding water to provide support.
 - C. fluid-filled compartments under pressure.**
 - D. external skeletons that provide support and protection against dehydration.
 - E. internal skeletons that are buried in soft tissue.
33. Galvanotaxis occurs when
- A. plants respond to gravity.
 - B. plants respond to hormones.
 - C. animals respond to temperature.
 - D. animals respond to metal ions.
 - E. protists respond to an electrical field.**
 - F. protists respond to touch.
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34. Pores on the leaf surface that function in gas and water exchange are called

- A. spongy mesophyll.
- B. xylem.
- C. phloem.
- D. waxy cuticles.
- E. stomata.**

35. (4 points) You knew it was coming...Write a *balanced* chemical equation for cellular respiration. In addition, include the molecule that is used to accomplish work in the cell.



Subtract 1 pt if one prefix number is missing or wrong (e.g., 4 H₂O, 2 pts if more than 1 is wrong.

Subtract 1 pt if one chemical formula is wrong (e.g., C₁₂H₆O₆), 2 pts if one missing or more than 1 wrong. Subtract 2 pt if missing ATP.

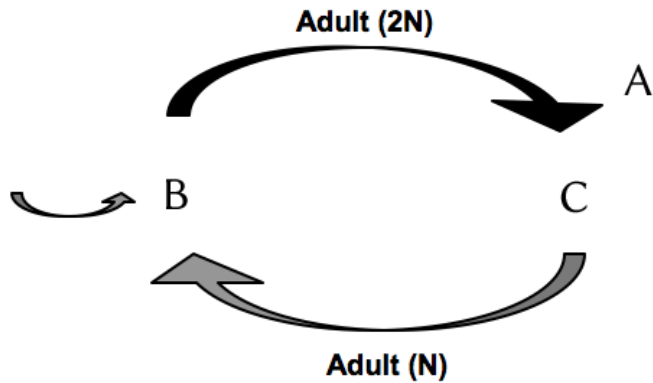
Ask me if uncertain.

Circle True or False. Read each statement completely before answering.

- 36. True **FALSE** A chemoheterotrophic protist is more likely to have an eyespot than a photoautotrophic protist.
- 37. **TRUE** False Adhesion is the hydrogen bonding of water molecules to the cell walls of the xylem.
- 38. True **FALSE** Plants spend most of their energy needs for photosynthesis.
- 39. **TRUE** False Open circulatory systems are less efficient than closed circulatory systems.
- 40. True **FALSE** High humidity increases the rate of transpiration.
- 41. **TRUE** False Without H⁺ ions bacteria could not move their flagella.
- 42. True **FALSE** Fungi and animals both spend most of their life cycle in the diploid stage.
- 43. True **FALSE** Plants seedlings display negative gravitaxis by growing away from gravity.
- 44. **TRUE** False Dynein is the motor protein found in eukaryotic flagella.
- 45. True **FALSE** A signal is not received at the organismal level in protists or bacteria.

Keep on keepin' on!

Use the following figure to fill in the blanks of the next four questions.



46. The process at A is **Meiosis**
47. The process (not the stage) that occurs at B is **Syngamy**
48. The stage at C (that results from A) is called **Spore**
49. Which of the three (A, B or C) does not exist in the animal lifecycle. **C**

Because they have to divide to multiply. Be wise, do good things, have a great summer!
